

Git for Scientists

So that we can still **prove what is human** when
working with AI in research.

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2023

<https://ghe-open.ch/slides/>

Digital Research in 2023

- Git was hard
- Manuscripts live in DOCX
- Track changes was the only record of who did what, and it vanishes on accept



Poll

Has AI written part of a document you produced this month?

👋 Hands up for yes 👋

Poll

Who can prove which parts of the document were generated by AI, and which parts by a human?

👋 Hands up for yes 👋

How do you prove it?

Ask yourself:

1. From memory?
2. Self documented?
3. Automated?
4. From the traces you leave?



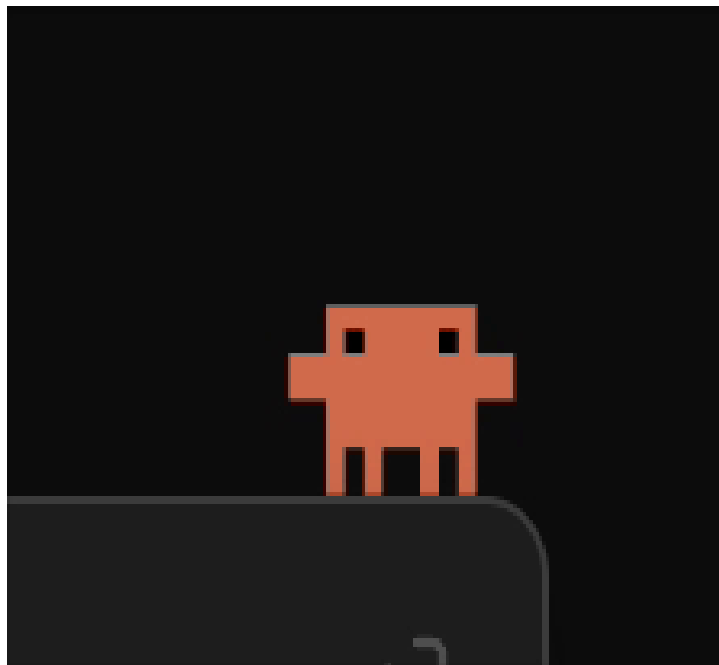
2026

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(Digital) Research in 2026

AI makes Git easier to use, and more necessary.

- Git is a language prompt away
- Tell the AI to track changes
- Plan well with GitHub issues
- Empower yourself with agentic AI
- Stay in control



Declaration of AI

International Science Council: A global reporting standard.

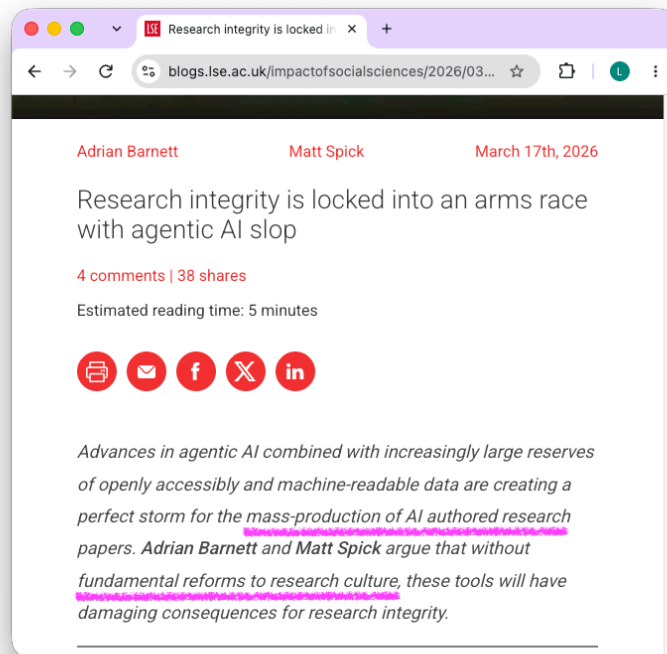
Proposed taxonomy

Based on the results of Round 1, a review of existing AI disclosure standards, and discussion within the core team, we propose 18 controlled categories (a “taxonomy”) to serve as a framework for disclosing AI use in research tasks. The proposed list of research tasks is:

1. *Idea generation*
2. *Hypothesis development*
3. *Literature search*
4. *Literature summarization*
5. *Design > Study design / experiment design*
6. *Design > Data collection methodology*
7. *Design > Data analysis plan*
8. *Data collection (operations / activities)*
9. *Data creation*
10. *Computer code generation/refinement/auditing*
11. *Data cleaning/preprocessing/auditing*
12. *Quantitative data analysis*
13. *Qualitative data analysis*
14. *Data visualization*
15. *Figure or image generation*
16. *Drafting manuscript or abstract text*
17. *Translation*
18. *Editing or rewriting including of reference lists*

Declaration of human

- A paper in 30 minutes ([Spick et al. 2026](#) )
- Well documented open data makes it possible
- How do we prove our work? ([Barnett and Spick 2026](#) )



Git for Scientists: teaching the need to know

So that we can prove what is human when working with AI in research.

- Read the tracked changes between two versions of a file (a diff)
- Understand how to experiment safely (a branch)
- Review what has gone into a set of changes (a pull request)
- Know what should never enter a repository, or an AI tool (data security)

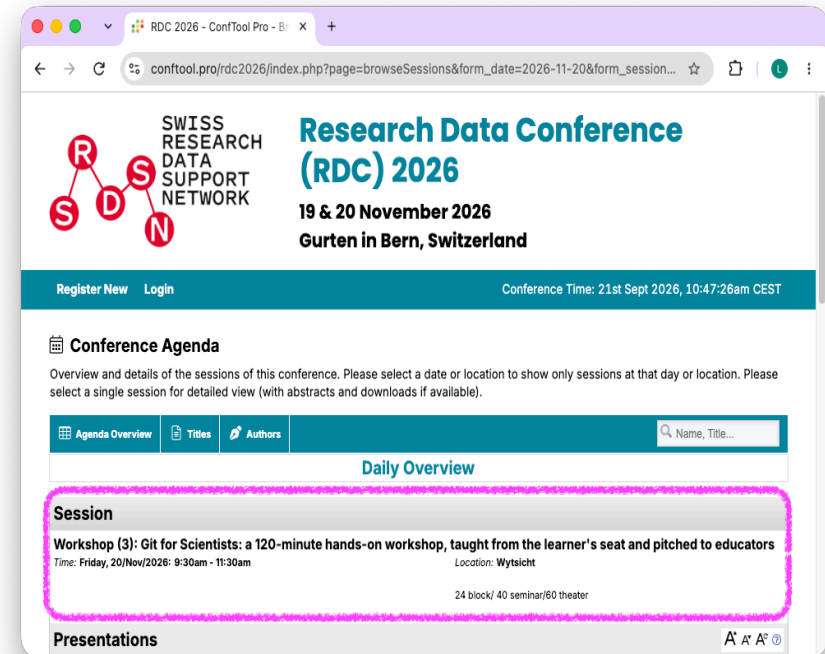
Git for Scientists: open educational resource

- Handbook for how to set up and run the workshop
- Open, CC BY, for any data steward or RSE to teach
- gitforsci.github.io/website 



Git for Scientists: RDC 2026, Bern

- Research Data Conference, 19 and 20 November 2026, Gurten, Bern
- 120-minute hands-on workshop, taught from the learner's seat and pitched to educators
- Friday 20 November, 9:30 to 11:30
- [Abstract](#)  and [ConfTool session](#) 



Agents for Scientists

<https://ghe-open.ch/slides/>[↑]

Agents for Scientists

Handing the need to know over to the AI

- **Work safely:** Git lets you revert
- **Experiment freely:** agentic AI opens endless possibilities
- **Track AI use:** prompts and resulting changes are recorded

First full-day workshop on 29 September.

FAIR by doing

- 15 participants, 12 months, 5 workshops
- Git, GitHub, FAIR data publishing, agentic AI
- One day with Greg Wilson: [Organizational Change for Open Science](#) ↗
- Anyone is welcome, especially qualitative researchers and people from non-IT backgrounds

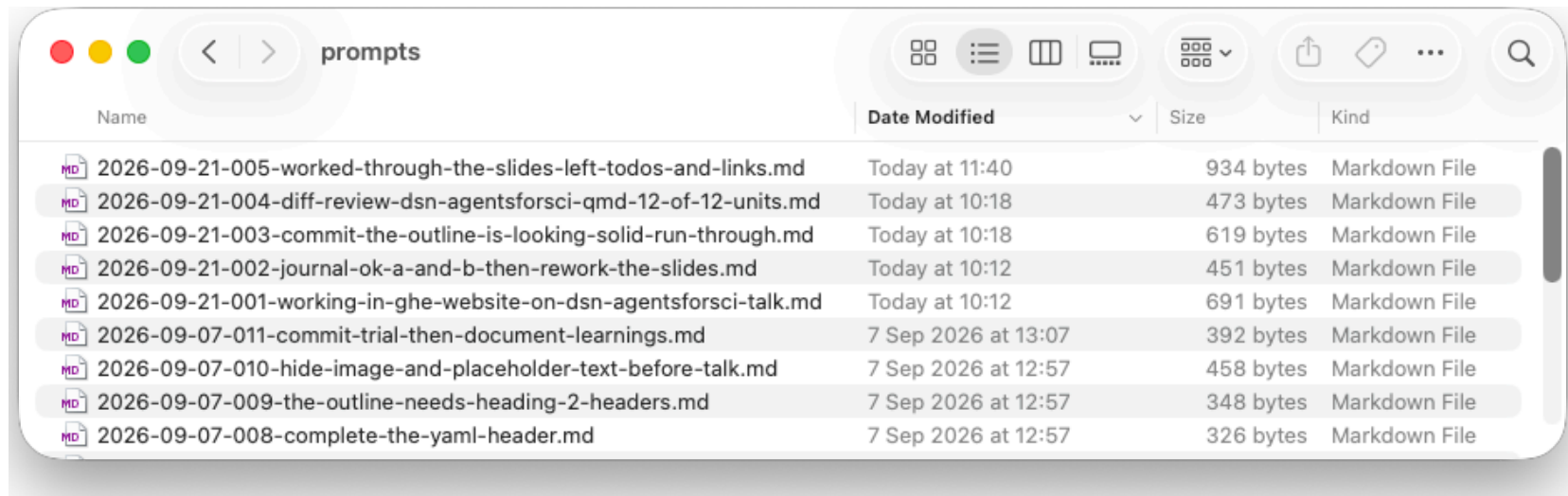
Express interest here:



Declaration of AI use

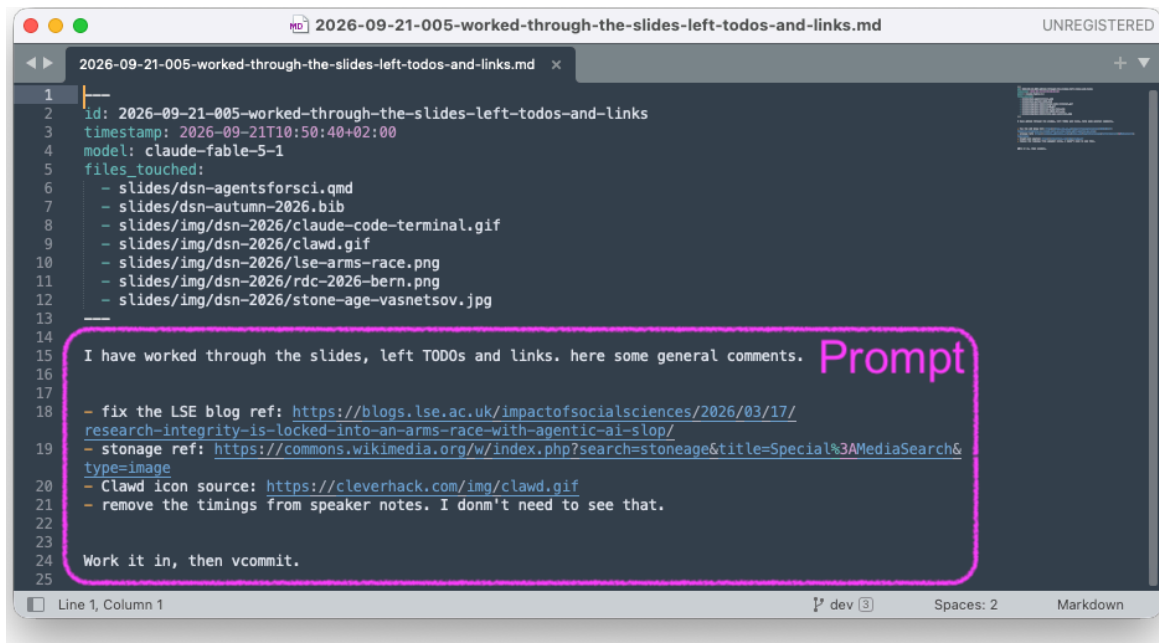
<https://ghe-open.ch/slides/>

A prompts folder in every repository



Every prompt recorded, one Markdown file each. This is the prompts folder of these slides.

One file per prompt

A screenshot of a code editor window. The title bar shows the file name '2026-09-21-005-worked-through-the-slides-left-todos-and-links.md' and 'UNREGISTERED'. The editor content is a markdown file with the following text:

```
1 ---
2 id: 2026-09-21-005-worked-through-the-slides-left-todos-and-links
3 timestamp: 2026-09-21T10:50:40+02:00
4 model: claude-fable-5-1
5 files_touched:
6   - slides/dsn-agentsforsci.qmd
7   - slides/dsn-autumn-2026.bib
8   - slides/img/dsn-2026/claude-code-terminal.gif
9   - slides/img/dsn-2026/clawd.gif
10  - slides/img/dsn-2026/lse-arms-race.png
11  - slides/img/dsn-2026/rdc-2026-bern.png
12  - slides/img/dsn-2026/stone-age-vasnetsov.jpg
13 ---
14
15 I have worked through the slides, left TODOs and links. here some general comments. Prompt
16
17 - fix the LSE blog ref: https://blogs.lse.ac.uk/impactofsocialsciences/2026/03/17/research-integrity-is-locked-into-an-arms-race-with-agentic-ai-slop/
18 - stonage ref: https://commons.wikimedia.org/w/index.php?search=stoneage&title=Special%3AMediaSearch&type=image
19 - Clawd icon source: https://cleverhack.com/img/clawd.gif
20 - remove the timings from speaker notes. I donm't need to see that.
21
22
23
24 Work it in, then vcommit.
25
```

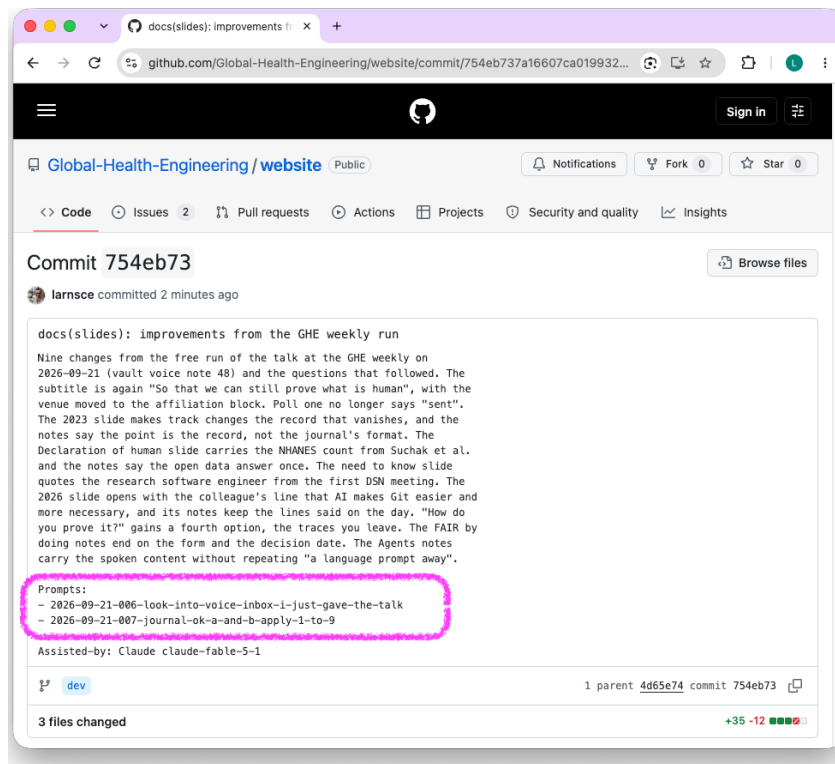
The text from line 15 to line 20 is enclosed in a pink dashed rectangular box. The word 'Prompt' is written in pink text to the right of line 15. The status bar at the bottom shows 'Line 1, Column 1', 'dev 3', 'Spaces: 2', and 'Markdown'.

An id and timestamp. The used AI model. The files that were touched.

<https://ghe-open.ch/slides/>

A commit skill

- Claude Code makes the commit
- Standardised
- The prompt files behind it are named
- Contribute at: [Global-Health-Engineering/skills](https://ghe-open.ch/slides/Global-Health-Engineering/skills) 



Let the AI declare your AI use

Prompt:

“For my latest DSN slides, use the traces of commits, prompt folder, pull requests, etc. to declare the use of AI against the [ETH Zurich Information Security and Artificial Intelligence \(AI\) Guide for Users of AI at ETH Zurich \(24 July 2025\)](#)  and the [ETH Library "Generative AI in the citation rules \(17 February 2025\)"](#) . Use only what's present in this repository.”

10 minutes: Get up and walk



The declaration, from the record

- **Tool:** Claude Code, model claude-fable-5-1, on 21 September 2026
- **Record:** 10 prompt files, 11 commits, 6 with an Assisted-by trailer
- **AI:** built the deck from outline and recording, notes, images, commits
- **Human:** outline, the talk twice, 26 review decisions, 5 own commits
- **Data:** public repository, CC BY 4.0, no personal or confidential data
- **Not recorded:** recordings, tool settings, ETH list of approved services

Full declaration: ghe-open.ch/slides/dsn-agentsforsci-ai-declaration 

Thanks

REFERENCES

- **These slides:** [download as PDF](#) 
- **AI use:** [declaration from the record](#) 
- Built with [Quarto](#)  and reveal.js
- **Section backgrounds:** image taken from [Danielle Navarro](#) 

Funding and support: [SwissRN](#)  local node ETH Zurich and the [Data Stewardship Network](#)  at ETH Zurich, funded by the [swissuniversities Open Science Programme](#) , and [Global Health Engineering](#) , ETH Zurich

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RESOURCES

- [Git for Scientists handbook](#) 
- [Agents for Scientists](#) 
- [FAIR by doing proposal](#) 
- [FAIR by doing: expression of interest form](#) 

References

- Barnett, Adrian, and Matt Spick. 2026. *Research Integrity Is Locked into an Arms Race with Agentic AI Slop*. LSE Impact of Social Sciences blog.
<https://blogs.lse.ac.uk/impactofsocialsciences/2026/03/17/research-integrity-is-locked-into-an-arms-race-with-agentic-ai-slop/> ↗.
- Spick, Matt, Tulsi Suchak, and Adrian Barnett. 2026. *The 30 Minute Manuscript: An Example Scientific Paper by AI Agent*. Octopus, preprint.
<https://doi.org/10.57874/cm47-c165> ↗.